

Alpha-synuclein, autophagy-lysosomal pathway, and Lewy bodies: Mutations, propagation, aggregation, and the formation of inclusions

Received for publication, April 2, 2024, and in revised form, August 21, 2024. Published, Papers in Press, September 2, 2024.

<https://doi.org/10.1016/j.jbc.2024.107742>

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Reviewed by members of the JBC Editorial Board. Edited by Phyllis Hanson

Research into the pathophysiology of Parkinson's disease (PD) is a fast-paced pursuit, with new findings about PD and other synucleinopathies being made each year. The involvement of various lysosomal proteins, such as TFEB, TMEM175, GBA, and LAMP1/2, marks the rising awareness about the importance of lysosomes in PD and other neurodegenerative disorders. This, along with recent developments regarding the involvement of microglia and the immune system in neurodegenerative diseases, has brought about a new era in neurodegeneration: the role of proinflammatory cytokines on the nervous system, and their downstream effects on mitochondria, lysosomal degradation, and autophagy. More effort is needed to understand the interplay between neuroimmunology and disease mechanisms, as many of the mechanisms remain enigmatic. α -synuclein, a key protein in PD and the main component of Lewy bodies, sits at the nexus between lysosomal degradation, autophagy, cellular stress, neuroimmunology, PD pathophysiology, and disease progression. This review revisits some fundamental knowledge about PD while capturing some of the latest trends in PD research, specifically as it relates to α -synuclein.

Parkinson's disease

Parkinson's disease (PD) is the second most common neurodegenerative disease, characterized by motor symptoms such as bradykinesia, dystonia, rigidity, postural instability, and resting tremors (1). A whole host of nonmotor symptoms, such as depression, anxiety, cognitive impairment, and several gastrointestinal-related impairments, also accompany PD (2). Motor symptoms in PD are mainly associated with the death of the pigmented dopaminergic (DA) neurons in the substantia nigra pars compacta (SNc), along with the formation of proteinaceous inclusions known as Lewy bodies (LBs; Fig. 1A). The primary recipient of the dopamine output of these neurons is the lateral striatum, located in the basal ganglia, whose lack of DA innervation is detrimental to "smooth" and "fluid"

movement (3–5). The lateral striatum, composed of the caudate and the putamen, is involved in assessing motor patterns, complex motor plans outputted by the cortex, and the selection of motor movements (6). The lateral striatum's inhibitory output is necessary for curtailing unnecessary movements and creating a smooth gait (Fig. 1B). With the lack of dopamine stimulation, the necessary inhibitory outputs of the striatum are abolished (Fig. 1C), leading to many of the motor symptoms seen in PD patients.

Ramifications for the basal ganglia

The DA input into the basal ganglia plays a role in two pathways: the direct and the indirect (Fig. 1B) (7). In the direct pathway, dopamine stimulates neurons in the lateral striatum equipped with D1 receptors. Excitatory cortical projections outputted from the glutamatergic neurons of the cortex also feed into the striatum. The lateral striatum, which contains inhibitory GABAergic neurons, then inhibits the globus pallidus internal segment (GPi) and the substantia nigra pars reticulata. Outputs of the substantia nigra pars reticulata and GPi are directed to the thalamus. The excitatory outputs of the ventral anterior and the ventral lateral regions of the thalamus are then directed to the frontal cortex, modulating the planning of movement. In the indirect pathway, inputs from the SNc stimulate the D2 receptors of the lateral striatum. Cortical glutamatergic projections also feed into this region. The lateral striatum then outputs an inhibitory signal to the external segment of the globus pallidus (GPe). Inhibitory projections from the GPe feed into the subthalamic nucleus, whose excitatory outputs stimulate GPi. GPi goes on to inhibit the thalamus, as it does in the direct pathway. Excitatory signals are then outputted from the thalamus to the motor cortex, responsible for the selection of movement. In healthy individuals, the direct and indirect pathways affect the planning and the selection of movements.

In the absence of the DA output from the SNc, due to degeneration observed in PD, the GABAergic neurons of the lateral striatum over-inhibit the GPe and do not sufficiently inhibit the GPi (Fig. 1C) (8). For the direct pathway, the lack of inhibition of the GPi leads to over-inhibition of the thalamus, resulting in less excitatory output to the cortex and impairing

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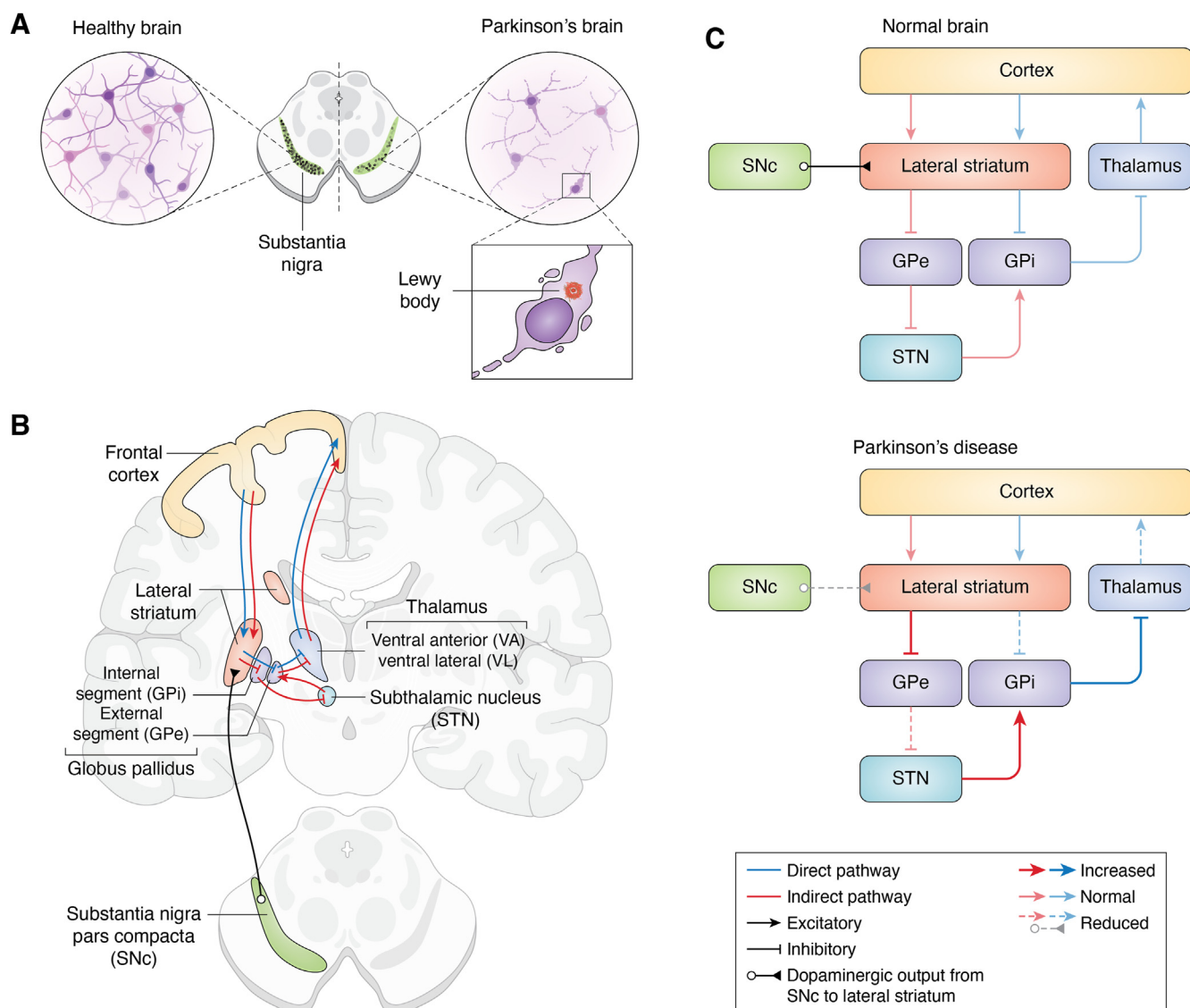


Figure 1. The pathological hallmark of PD and the downstream effects of the SNc's dopaminergic output into the basal ganglia. *A*, postmortem samples show the loss of pigmented DA neurons in the SNc. Healthy patients show a high density of the melanin-filled DA neurons in the substantia nigra, while PD patients exhibit the loss of cells in this region. Upon closer inspection, many of the remaining DA neurons in PD patients show cytoplasmic inclusions, referred to as Lewy bodies. *B*, DA neurons of the substantia nigra release dopamine to stimulate and inhibit nuclei located in the lateral striatum, which include the caudate and the putamen. Both pathways require DA input from the SNc to modulate the planning and selection of motor actions appropriately. The dopaminergic input into the basal ganglia is shown in black. Blue lines represent the direct pathway, and the red lines represent the indirect pathway. *C*, consequences of Parkinson's disease on the basal ganglia pathways. With adequate dopaminergic stimulation of the lateral striatum via DA projections, both the direct and indirect pathways can modulate the planning and selection of movement. However, the loss of dopaminergic input results in decreased activity of the direct pathway and increased activity of the indirect pathway, which culminates in the lack of thalamic input to the motor cortex. DA, dopaminergic; PD, Parkinson's disease; SNc, substantia nigra pars compacta.

motor planning. For the indirect pathway, the lack of inhibition of the subthalamic nucleus leads to overstimulation of the GPi, which in turn over inhibits the thalamus, crippling its excitatory signal to the cortex, impairing the selection of movement. The impairments of these pathways explain the rigidity and bradykinesia symptoms of PD.

Beyond motor symptoms

Although alleviating the motor symptoms of PD has been the central focus of clinical research, more focus and research need to be done on the underlying cause of these motor symptoms. The degenerative mechanism first initiated

by the death of DA neurons in SNc continues to spread and affect neurons through the central nervous system (9). The hallmark of this degeneration is proteinaceous inclusions, which are also present in Lewy body dementia (LBD), another neurodegenerative disease with mainly cognitive impairments and loss of memory (10). In sum, while treating the motor symptoms of PD is essential to improve quality of life, it is more important to understand the mechanism involved in neuronal degeneration. Due to it being a major constituent of LBs and its aggregation being present in all synucleinopathies (11), α -synuclein (α -syn) will be the focus of this review.

Alpha-synuclein

With the discovery that alpha-synuclein (α -syn) is a significant component of LBs (12), coupled with its role in familial Parkinson's disease (13), both in its mutated and overexpressing form (14–18), α -syn has become the central protein in PD research, and one of the most enigmatic proteins in biology. With its 14 kDa size, composed of 140 amino acids, it is hard to imagine how this protein has remained a mystery for so long. Part of this mystery stems from its intrinsically disordered nature (19), with its native structure and form being hard to ascertain. Moreover, it has also been posited that physiologically, α -syn may function as a multimeric protein (20–23). α -syn's cellular localization and function are somewhat diverse, as it resides in the cytoplasm physiologically but has an affinity toward membranes (24). Research in recent years has found that α -syn interacts with anionic phospholipids, associates with membrane curvature, and is involved in the organization and tethering of synaptic vesicles (25, 26), modulating synaptic vesicle release (27),

vesicle recycling (28), and mediating SNARE-complex assembly (29). Its ability to interact with membranes is enabled through its seven imperfect KTKEGV repeats (Fig. 2A) (30). Under disease conditions, α -syn accumulation has been shown in the cytosol, endolysosomal system, exosomes, and even the nucleus (24, 31–34). Its ability to aggregate is primarily due to its nonamyloid component (NAC), which is hydrophobic (35). A detailed structure accompanied by labeling of domains, regions, repeats, and mutations of the α -syn are illustrated in Figure 2A.

Although α -syn's association with membranes is undeniable, few have suggested that these membrane-associated functions of α -syn are only one aspect of its functions in the cell. Recent research indicates that α -syn also plays a role in cellular stress, cellular homeostasis, and mRNA stability (36–40). α -syn levels are indicators of cellular health when it comes to autophagy (41–44), specifically chaperone-mediated autophagy (CMA) (45–47). Stress signaling (such as immune activation) and exposure to toxic chemicals have also been

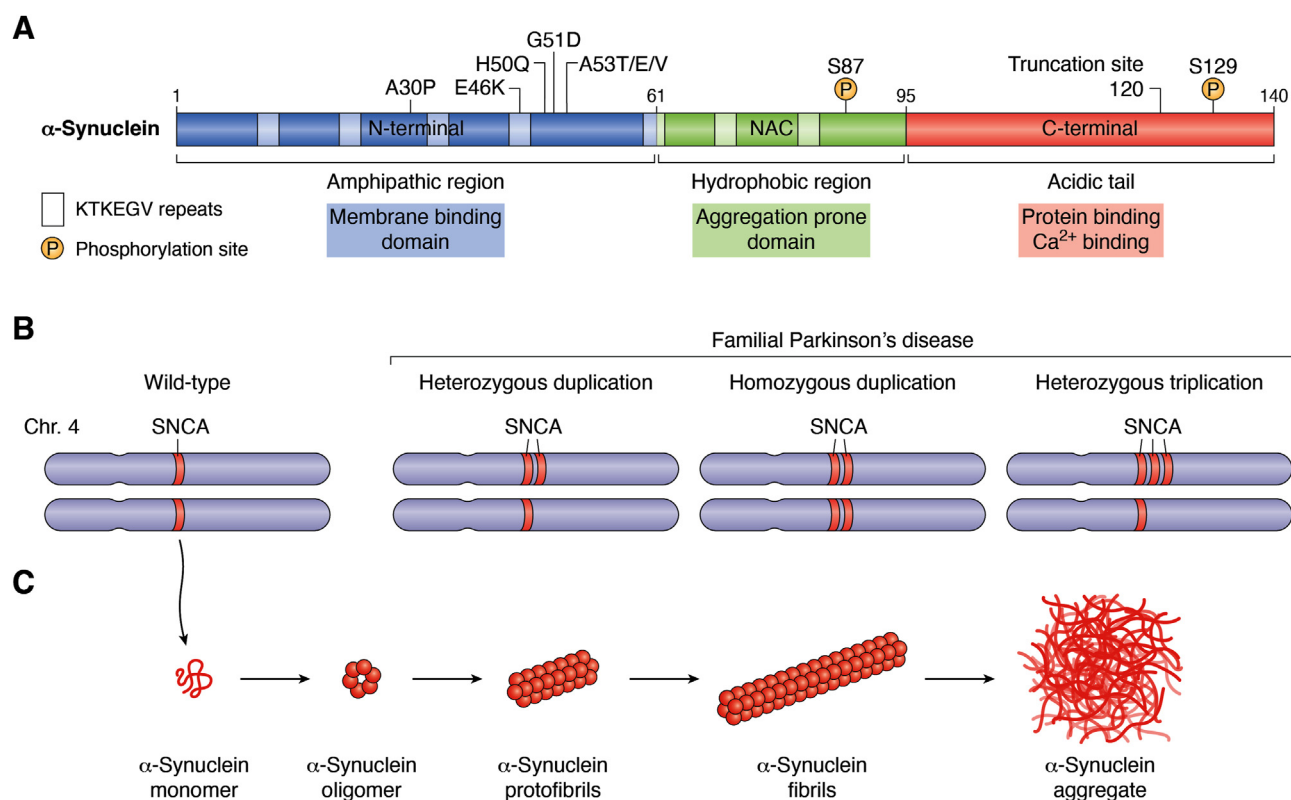


Figure 2. α -syn protein structure and important regions, SNCA multiplication, and aggregation of α -syn. A, α -syn, a 14 kDa protein with 140 amino acids, contains three domains: The membrane-binding domain, the NAC (nonamyloid component) region, and the protein-binding domain. The membrane binding domain, which is on the N-terminal side of the protein, contains most of the KTKEGV (imperfect) repeats that aid in membrane binding. The NAC domain is the hydrophobic region that is critical for the aggregation of α -syn monomers. The C terminal contains the protein-binding domain, which includes the most widely known phosphorylation site: serine 129. Familial PD, caused by mutations in SNCA, translates to the variants detailed in the membrane-binding region of the protein. These mutations include the following: A53T, A53E, A53V, A30P, H50Q, G51D, and E46K. B, patients with familial cases of PD, due to multiplication of the SNCA locus (located on chromosome number 4), have one additional copy of the SNCA gene compared to WT when an extra copy is inherited from one parent (i.e., heterozygous duplication). In cases where additional copies of the gene are inherited from both parents, patients have two additional copies of the SNCA gene. Such cases are referred to as homozygous duplication. Finally, patients known to have a heterozygous triplication of the SNCA gene inherited three copies of the SNCA gene from one parent and only one copy of the SNCA gene from another parent. Theoretically, homozygous duplication and heterozygous triplication should express similar amounts of α -syn, but epigenetic factors and allele preferences create a lot of variety in protein expression. Both homozygous duplication and heterozygous triplication are associated with earlier onset of PD. C, α -syn monomers have been shown to aggregate into oligomers and protofibrils before their fibrillization. Protocols have been established for the generation of α -syn monomers, along with protocols to produce full-length fibrils (larger than 100 μm) or preformed fibrils (smaller than 100 μm), usually made from the sonication of full-length fibrils into smaller chunks (250–252). PD, Parkinson's disease; SNC, substantia nigra pars compacta.

shown to upregulate α -syn (48–50). Increased α -syn expression has also been associated with mitochondrial stress, higher levels of oxidative species and even endoplasmic reticulum stress (51–54). Lastly, α -syn’s clear ability to interact and modulate processing bodies, organelles that play a role in mRNA turnover and storage, adds another facet to our understanding of α -syn’s function in the cell (40). For all these reasons, it is clear that α -syn homeostasis is paramount to maintaining healthy neuronal function (55). These functions suggest that although membrane association and synaptic vesicle trafficking might be some of α -syn’s roles in the cell, they are certainly not the only functions this protein serves. Table 1 lists some of the more recent findings about the physiological functions of α -syn.

α -syn in Lewy bodies

Under disease conditions, especially in synucleinopathies, α -syn is known for its accumulation and aggregation (11). α -syn’s accumulation in cytoplasmic inclusions, such as LBs and glial cytoplasmic inclusions (56), have earned α -syn its “toxic” and “pathological” labels. Perhaps the most crucial question to be addressed is whether α -syn is a symptom or the reason behind the cellular aggregates/sequestration processes. Most would argue that α -syn is essential for the pathophysiology and the mechanisms driving inclusion formation and that its aggregation is linked to the formation of inclusions through tethering and pulling together of cellular organelles, stabilizing them in inclusionary bodies in the cytoplasm, which go on to mature into LBs (57–59), in a cell-autonomous manner. However, fibrillization may not even be necessary (60) in driving the formation of LBs, but its oligomerization and accumulation lead to downstream effects such as oxidative stress and impaired organelles (36, 61, 62). This, coupled with dysfunctional autophagy (43), degradation (45, 63, 64), and

exposure to external toxic chemicals (61), may lead to the onset and progression of PD.

α -syn in familial PD

Genetic data involving familial PD cases suggest that α -syn plays a more fundamental role in the formation of cellular inclusions, as individuals with mutations in *SNCA*, the gene encoding α -syn, exhibit an autosomal dominant inheritance of PD (65). Two types of modifications of the *SNCA* locus have become central in PD research: missense point mutations and locus multiplication. The first type of mutation, missense mutations, have been shown to occur across different families (66), with the A53T mutation being the first to be discovered and garnering the most attention (13). Common among most of these missense mutations in the *SNCA* locus is that they enhance the aggregation ability of α -syn (67). Additionally, most of these mutations occur near the end of the membrane-binding region of the protein (Fig. 2A), immediately preceding the aggregation-prone NAC region. These misfolded monomers then allow the formation of aggregates, which then act as prions to enable the misfolding of other α -syn monomers and, therefore, serve as a seed for further aggregation of α -syn (68, 69).

The second type of modification of the *SNCA* locus is *SNCA* duplication and triplication (16–18) (Fig. 2B). In these familial cases, patients present with extra copies of the *SNCA* locus in their genome, driving higher expression levels of α -syn in cells. These cases show another side of α -syn aggregation: higher concentrations of α -syn increase the likelihood of its aggregation (70, 71). This relationship between α -syn concentration and aggregation has been well-established *in vitro* and *in vivo* models (59, 72, 73).

The missense mutations and multiplication of *SNCA* have cemented the role of α -syn aggregation in PD. One key question is why patients with *SNCA* mutations, with cells expressing misfolded and aggregate-prone α -syn variants, or those expressing higher concentrations of α -syn, present with PD-related symptoms in middle age or later in life. Although patients with missense mutations usually exhibit early onset PD (2) (*i.e.*, PD diagnosis occurring before the age of 50), it is still puzzling why it takes many years for higher concentrations and misfolded forms of α -syn to form aggregates, disrupting neuronal function and initiating neurodegeneration. *In vitro*, α -syn variants have been shown to quickly form oligomers and initiate fibrillization (74–78). The disparity between the *in vivo* and *in vitro* time scales needs to be studied further. Additionally, α -syn aggregation alone seems to be not associated with the onset of the motor symptoms of PD but is associated with prodromal PD (79–81), as prodromal patient’s cells contain α -syn aggregates; yet, patients do not present with PD motor symptoms. This has led some to explore a secondary hit, an additional insult that must be present for the manifestation and onset of PD motor symptoms to occur (82, 83).

Finally, it is important to note that more prevalent and penetrant forms of PD-related mutations exist, such as *GBA*, *LRKK2*, *PRKN*, and *PINK1* (84), which provide another view to

Table 1
Physiological functions of α -syn

Function of α -syn	Reference
Involved in synaptic vesicle exocytosis, through modulating the assembly of the SNARE-complex	(29)
Detection and formation of membrane curvature	(246)
Highly abundant and localized to the synaptic bouton	(247)
Attenuates neurotransmitter release and synaptic vesicle clustering	(28)
Regulation of synaptic vesicle endocytosis	(248)
Tethering of synaptic vesicles, determinant of synaptic size, and involved in synaptic vesicle clustering	(26)
Interaction with endocytic trafficking, retromer complex, and mRNA binding proteins	(249)
Association with calcium allow for synaptic vesicle binding at the C terminal	(250)
Modulation of exocytosis, binding to membranes, association with mitochondria	(251)
Enables endocytosis through upregulation of phosphatidylinositol 4,5-bisphosphate, mutations alter clathrin-mediated endocytosis	(252)
Modulation of dopamine release in substantia nigra neurons—in an action potential related manner	(253)
Regulation of cellular iron homeostasis	(254)
Docking of synaptic vesicles to the presynaptic membrane through lipid links	(255)
Involved in clathrin assembly	(256)
Regulation of iron metabolism	(257)
Modulates processing bodies along with mRNA stability	(40)

our understanding of PD pathophysiology. Many patients with *LRKK2* mutations and all patients with *PRKN* and *PINK1* mutations do not present with Lewy pathology while exhibiting all the symptoms of PD. However, since idiopathic PD comprises more than 85% of cases in PD (85), most of which present with LBs (86), the focus on α -syn-related aggregation has become one of the main avenues of inquiry into PD.

α -syn propagation

One reason why PD, whether idiopathic or familial, is more prominent in the elderly and has a long prodromal period might be due to α -syn propagation; disease progression and α -syn propagation may, therefore, be linked (87). Many have emphasized the role of α -syn propagation, allowing aggregated or misfolded forms of α -syn originating from one neuron to propagate and seed the aggregation of endogenous α -syn in neighboring neurons. These are said to follow connectivity pathways in the brain (88). Although this process has been observed rapidly *in vitro*, partially facilitated by a smaller number of cells and higher concentrations of exogenous α -syn, this process may take years to occur *in vivo*. While some have suggested their disbelief in the role of the prion-like transfer of α -syn in PD (89), others have maintained that it is a crucial and central event in PD (90–92).

α -syn oligomers

If α -syn propagation plays a central role in PD, it is crucial to find precisely what form of α -syn is involved in propagation (Fig. 2C), a hotly debated topic in α -syn research (93). For many years, the fibril form of α -syn was thought to be responsible for propagation due to its ability to seed α -syn aggregation in cells once internalized. More recently, protofibril and oligomeric forms of α -syn have gained prominence for their potential to act as the propagating form of α -syn due to their ability to be quickly internalized while maintaining the ability to seed aggregates (94). Also, some point mutations in *SNCA* enhance only oligomerization but not fibrillization (95); this suggests that fibrillization may not be a significant factor in disease progression and that the fibril forms of α -syn is not the most “toxic” form. Whether soluble or insoluble forms of α -syn are involved in seeding and aggregate formation in PD will need to be investigated further. These studies must ascertain which form of α -syn can be found early in PD and is, therefore, involved in its onset.

Strains of α -syn

Further complexities are associated with α -syn oligomers and fibrils, as different strains of α -syn aggregates have different branching characteristics, different thioflavin T fluorescence, and different cell internalization and aggregation properties. In one study, the fibrillization of α -syn in the presence and absence of NaCl resulted in markedly different fibrillized products (96). These strains then targeted different brain regions *in vivo*, with a propensity to target specific cell types (97). Strains went on to form conformationally different intracellular phospho- α -syn

aggregates. Similarly, the establishment of ribbon and fibrillar strains of α -syn led to strain-dependent propagation in another *in vivo* study (98). These findings further complicate research into α -syn propagation; however, research into the strains of α -syn can potentially explain the differential nigral and striatal pathologies, along with morphologically different aggregates/inclusions that form in various synucleinopathies (99, 100).

α -syn endocytosis

The next important question in the propagation of α -syn is how it is internalized into the cell (Fig. 3). For many years, researchers believed that clathrin-mediated endocytosis (CME) was involved in the uptake of α -syn (101–105), with some articles even proposing that the internalization mechanism is receptor-mediated, the most known and somewhat controversial receptor being lymphocyte-activation gene 3 (106, 107). Although some articles carefully distinguish between dynamin-mediated internalization and CME (108), most articles conflate dynamin-mediated inhibition of α -syn uptake to suggest that α -syn internalization occurs through CME. As discussed in our previous article (109), dynamin inhibition can affect multiple endocytic pathways (110–114). Additionally, inhibitors such as dynasore and Pitstop have many off-target effects (115, 116), making them a suboptimal option for selective inhibition of cargo internalization. Multiple other pathways for α -syn internalization have been proposed such as the macrophage-specific phagocytosis (117), lipid raft-mediated endocytosis (118), and caveolae-dependent internalization (119). The lack of specific inhibitors, the employment of silencing technology to inhibit CME directly, and the various pathways suggested in the internalization of α -syn led us to study this process ourselves (109).

α -syn macropinocytosis

Macropinocytosis is an actin-mediated endocytic mechanism that allows for extracellular fluid uptake (120). Macropinocytosis is preceded by the ruffling of the plasma membrane and has been shown to play a major role in the uptake of dextran and large exogenous materials such as viruses (111, 121, 122). Previously, very few articles discussed the macropinocytic entry of α -syn, with only one article showing that macropinocytosis may play a role (123), while another article reported the ability of α -syn to induce membrane ruffling, but its internalization was not inhibited by ethyl-isopropyl amiloride (124), a macropinocytic inhibitor. Much of the confusion arises from the many different forms of α -syn used to study its internalization (*i.e.*, monomeric, oligomeric, protofibril, and fibril) and the lack of specific cell biological assays to assess internalization. Using the genetic approach of clathrin heavy chain knockdown, coupled with following α -syn internalization *via* EM, we found no evidence for CME regarding α -syn preformed fibrils (PFFs) uptake (109). Actin-mediated membrane ruffling, the recruitment of Rac1, and the engulfment of α -syn PFFs by membrane ruffles captured *via* EM provide convincing evidence for the macropinocytic

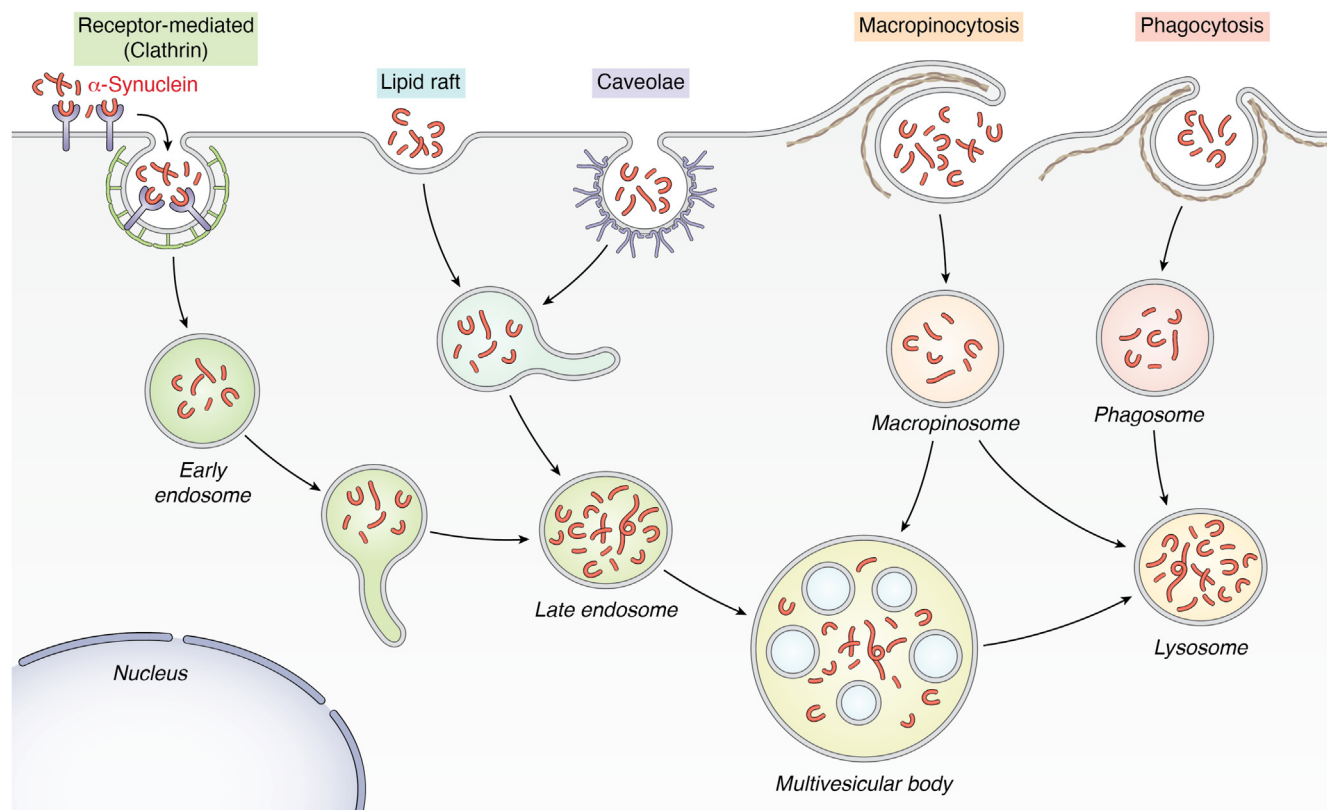


Figure 3. α -syn internalization pathways. Multiple pathways have been proposed for the internalization of α -syn monomers, oligomers, and fibrils. Here, we illustrate five of the most reported pathways: receptor-mediated (CME), lipid raft, caveolae, macropinocytic, and phagocytic internalization pathways. Cargo internalized through CME, lipid raft, and caveolae go through the endosomal system, undergo endosomal maturation, and are then transported to lysosomes. Although traditional macropinocytosis also colocalizes with markers of early and late endosomes, the type of macropinocytic pathway we discovered did not undertake this maturation, allowing cargo within macropinosomes to be quickly transported to lysosomes. Finally, phagocytosis, an internalization mechanism specific to macrophages, has been shown to take up α -syn, be transported into phagosomes, and eventually be transported to lysosomes. Conventional phagocytic internalization also colocalizes with early endosome markers and undergoes maturation before reaching lysosomes. CME, clathrin-mediated endocytosis.

uptake of PFFs. Since publishing our findings, others have corroborated our results in another study (125).

α -syn exocytosis

Another area of research regarding α -syn propagation focuses on its exocytosis. One hypothesis suggests that membrane leakage caused by cellular apoptosis or necrosis releases aggregate forms of α -syn, which are then available for uptake by neighboring cells (126). Others have suggested regulated pathways in which α -syn can be transported to other cells (127). The apoptosis model of α -syn spread has remained less convincing since *in vitro* α -syn overexpression and internalization models have rarely resulted in cellular apoptosis in our hands. In fact, many studies that study the intracellular and intercellular transport of α -syn have failed to report significant changes in lactate dehydrogenase levels (128–131). Additionally, while degeneration is rampant in DA neurons of the SNc, this degeneration takes decades, as does the onset of PD motor symptoms, while α -syn aggregates continue to spread during the prodromal period (132, 133).

Exosomal release of α -syn internalization has garnered the most support in studying the intercellular transfer of α -syn aggregates (44, 134–138). Still, a few have suggested tunneling

nanotubes may play a role in this process (139–142). In our previous study of α -syn internalization and trafficking, we inquired about the exocytic transport of α -syn PFFs (109). We observed that α -syn PFFs, previously internalized, are then transported to other cells *via* extracellular vesicles/exosomes, with α -syn PFFs residing on the surface of exosomes and not within their lumen. Furthermore, we found that internalization of α -syn exosomes was inhibited using the macropinocytic inhibitors, previously used to inhibit the initial internalization of PFFs into the cell, suggesting that α -syn aggregates may continue to play a role in their own propagation even after their internalization by cells, due to their position on the surface of exosomes. Table 2 contains a collection of previous publications that studied the internalization and/or intercellular transmission of α -syn.

Lysosomal membrane permeabilization

One aspect of α -syn propagation that has been difficult to ascertain is how intercellularly propagating α -syn aggregates seed endogenous α -syn in their intercellular journey. A closer look into lysosomal membrane permeabilization and the recruitment of galectin-3 used as a marker for ruptured vesicles/vacuoles (143), has provided a potential answer (144).

Table 2**Internalization and intercellular transfer of α -syn**

Findings	Reference
Rab5A-mediated uptake of monomeric α -syn was observed.	(258)
Cytochalasin D treatment led to decreased fibril α -syn uptake in microglia.	(36)
Coimmunoprecipitation of clathrin and α -syn aggregates; colocalization of aggregated α -syn with clathrin in microglia.	(101)
Microglia can internalize and degrade α -syn aggregates; accumulation occurred upon activation of microglia.	(259)
Internalization of α -syn fibrils was inhibited by low temperature and dynamin dominant-negative. Monomeric α -syn entered cells through diffusion and colocalized with EEA1 and LAMP2.	(260)
Internalization of α -syn monomers not inhibited with dynamin dominant-negative nor with dynasore, but lipid-raft mediated.	(118)
α -syn transmission was inhibited with the use of dynamin dominant-negative; using an endogenous expression model.	(261)
α -syn endocytosis was inhibited in cells expressing dynamin dominant-negative; using an endogenous expression model.	(262)
α -syn endocytosis and transfer occur in a dynamin-dependent manner.	(263)
Monomeric α -syn endocytosis was decreased by inhibiting dynamin GTPases.	(264)
Intercellular transfer of α -syn from rat brain to graft DA neurons. The transferred α -syn colocalize with EEA1. Nonaggregated forms of α -syn can transfer from host to graft.	(265)
Administered together, tau, TAT, and α -syn colocalize, indicating α -syn fibrils are internalized using macropinocytosis.	(123)
Dynasore inhibited monomeric, oligomeric, and fibril α -syn uptake in a concentration-dependent manner.	(266)
Monoclonal antibodies targeting α -syn inhibit its internalization and intercellular transfer.	(267)
α -syn fibrils induced membrane ruffling, but ethyl-isopropyl amiloride, a macropinocytic inhibitor, did not inhibit its internalization.	(124)
Clathrin-mediated internalization of α -syn, using an endogenous expression model and intercellular transfer of α -syn. Colocalization of α -syn with EEA1, NR1, and NR2A.	(102)
Oligomeric α -syn's intercellular trafficking is elevated following knockdown of Rab8b and Rab13. Endocytic recycling of α -syn relies on Rab11a and Rab13; overexpression leads to increased secretion.	(268)
Dynamin is partially required for the uptake, but not the intercellular transfer of α -syn fibrils. Colocalization of the fibrils with LAMP1.	(138)
Intercellular spread of α -syn relies on tunneling nanotubes.	
Inhibition of clathrin-mediated endocytosis using dynasore resulted in decreased uptake of α -syn fibrils and phospho-fibrils.	(103)
LAG3 mediated endocytosis of α -syn PFF: receptor-mediated endocytosis. Colocalization with Rab5, Rab7, and LAMP1.	(106)
PFFs are transported through the endolysosomal pathway and colocalizes with LAMP1.	(269)
DJ-1 knockdown reduces phagocytic uptake of soluble α -syn by microglia.	(270)
GGA3 contributes to endosomal oligomerization of α -syn and drives its secretion.	(271)
Serine protease KLK6, along with proMMP2, can cleave extracellular fibril α -syn, inhibiting its propagation. KLK6 deficient showed increased α -syn uptake.	(272)
α -syn aggregates depend on heparan sulfate for internalization.	(35)
<i>In vivo</i> injection of phosphorylated fibrils resulted in more toxicity and neuronal loss compared nonphosphorylated fibrils. S129 phosphorylation enhanced α -syn fibril uptake by neurons.	(273)
Exosome associated-oligomeric α -syn does not rely on heparan sulfate proteoglycans for internalization, and its internalization is much higher than naked α -syn oligomers.	(274)
α -syn monomers use dynamin-mediated endocytosis, colocalize with Rab 4, 5, and 7. α -syn fibrils undertake a different pathway.	(108)
Ubiquitin C-terminal hydrolase L1 is involved in the intercellular transmission of α -syn.	(275)
LAG3 and RAB5A found to be genetic predictors for regional brain atrophy. Highly expressing LAG3 regions of brain correlated with greater atrophy.	(276)
Fc γ RIIB binds α -syn fibrils and mediated intercellular transmission through its signaling.	(277)
Oligomeric α -syn binding and colocalization with connexin-32, during its uptake. Expression of connexin-32 was linked to the intercellular transfer of α -syn. Inhibitors of connexin-32 reduce α -syn uptake.	(278)
TRPML1 activation resulted in the upregulation of lysosomal exocytosis, reducing the burden of α -syn accumulation.	(279)
CME of aggregated α -syn: colocalization of α -syn with transferrin, EEA1, LAMP1, LAMP2, and Rab11. Aggregated α -syn shows higher cellular accumulation than monomeric α -syn.	(104)
Rab27b is involved in the release and clearing of α -syn, using a doxycycline-inducible α -syn expression.	(280)
Cellular entry of α -syn fibrils is facilitated through a myosin-7B-dependent pathway.	(281)
Caveolae-dependent uptake of α -syn in dopaminergic neurons was observed.	(119)
Synaptic transmission of α -syn is bidirectional, mainly occurring in retrograde. Retrograde trans-synaptic uptake of α -syn occurred through an endophilin-independent manner.	(282)
α -syn fibrils are intercellularly transferred through tunneling nanotubes.	(142)
Large oligomers are dependent on clathrin-mediated endocytosis and colocalize with early endosomes and lysosomes.	(105)
LAG3 D1 and APLP1 E1 domains bind to α -syn's c terminal. Phosphorylation of α -syn at S129 enhances its internalization.	(283)
Sequestration of LC3 by mutated endogenously expressed α -syn, stresses the degradative/autophagy pathway, facilitating exocytosis of α -syn.	(284)
α -syn PFFs were found to colocalize with LysoTracker.	(285)
Endogenously expressed α -syn undergoes caveolin-dependent endocytosis and colocalizes with EEA1 and Rab7.	(286)
Lysosomes release nonmembrane-bound α -syn aggregates from neurons.	(256)
Heparan sulfate proteoglycans mediate α -syn PFFs toxicity. Knockdown of proteins responsible for heparan sulfate proteoglycan synthesis was protective against degeneration.	(287)
α -syn fibrils use actin-mediated macropinocytosis to be internalized into cells. No significant involvement of caveolae-mediated endocytosis and clathrin-mediated endocytosis was found.	(125)
α -syn PFFs use a rapid form of macropinocytosis to be internalized into cells and be transferred to lysosomes. PFFs are transported on the surface of exosomes to neighboring cells.	(109)
Antibodies targeting the c terminal of α -syn, block PFF uptake; decreased phospho-synuclein accumulation.	(288)

Lysosomal membrane rupture can occur due to cellular stress (145). Although we have not found PFFs alone to induce lysosomal leakage, others have (146, 147). This would allow lysosomal contents to meet cytosolic proteins, allowing α -syn aggregates to seed the aggregation of endogenous α -syn. Several articles have shown the mechanisms underlying lysosomal membrane permeabilization and galectin-3 recruitment (147). Another important question to note: how is this level of lysosomal dysfunction induced, and how does it occur in the

brains of PD patients? In our recent study (148), we show direct evidence for the leakage of lysosomal contents, specifically PFFs, into the cytosol caused by the combined exposure of α -syn PFFs and interferon-gamma (IFN- γ),

α -syn inclusions in synucleinopathies

The discussion regarding forms of α -syn, its endocytosis, and propagation are all important steps that lead to the

eventual consequence of α -syn accumulation and aggregation in PD: formation of LBs. Whether α -syn, in its different forms, plays a mechanistic role in initiating LBs or is incorporated in LBs is debated; however, the presence of α -syn in LBs is not. α -syn has been found in LBs repeatedly since its original localization (12). However, α -syn aggregation is not specific to PD and is present in other synucleinopathies, most notably LBD, multiple system atrophy and Krabbe disease (10, 149, 150). In LBD, α -syn accumulates in LBs as in PD; however, LBD differs symptomatically from PD due to the initial neurodegeneration also occurring in the neocortex rather than just the substantia nigra, as is the case in PD (151–153). In multiple system atrophy, α -syn accumulates in inclusions termed Papp-Lantos bodies (154), which occur primarily in glial cells. Krabbe disease, a rare case of infantile neurodegenerative disease, is a disease in which α -syn aggregation and inclusion formation occur in infant brains (155). Due to the prevalence of PD and LBD, LBs have garnered more attention than α -syn-positive

inclusions occurring in other synucleinopathies; however, even in the case of LBs, the field is still lagging in understanding their structure, origin, and formation.

α -syn aggregates and inclusions look different across different synucleinopathies. This may be due to the different strains of α -syn aggregates formed. Environmental differences, such as the presence of NaCl, have drastic effects on the aggregation of α -syn (96). Cryo-EM and NMR structures of α -syn fibrils/filaments formed *in vitro* using patient aggregate seeds have exemplified how different α -syn aggregates can be. Some examples of the different *in vitro* aggregates formed from different synucleinopathy seeds or using patient cerebrospinal fluid to feed α -syn aggregation have been shown in Figure 4. It is important to note that recent research has found that the fibrils formed through amplification do not have the same 3D structure as the original seeds used (156); clearly, this disparity warrants further investigation.

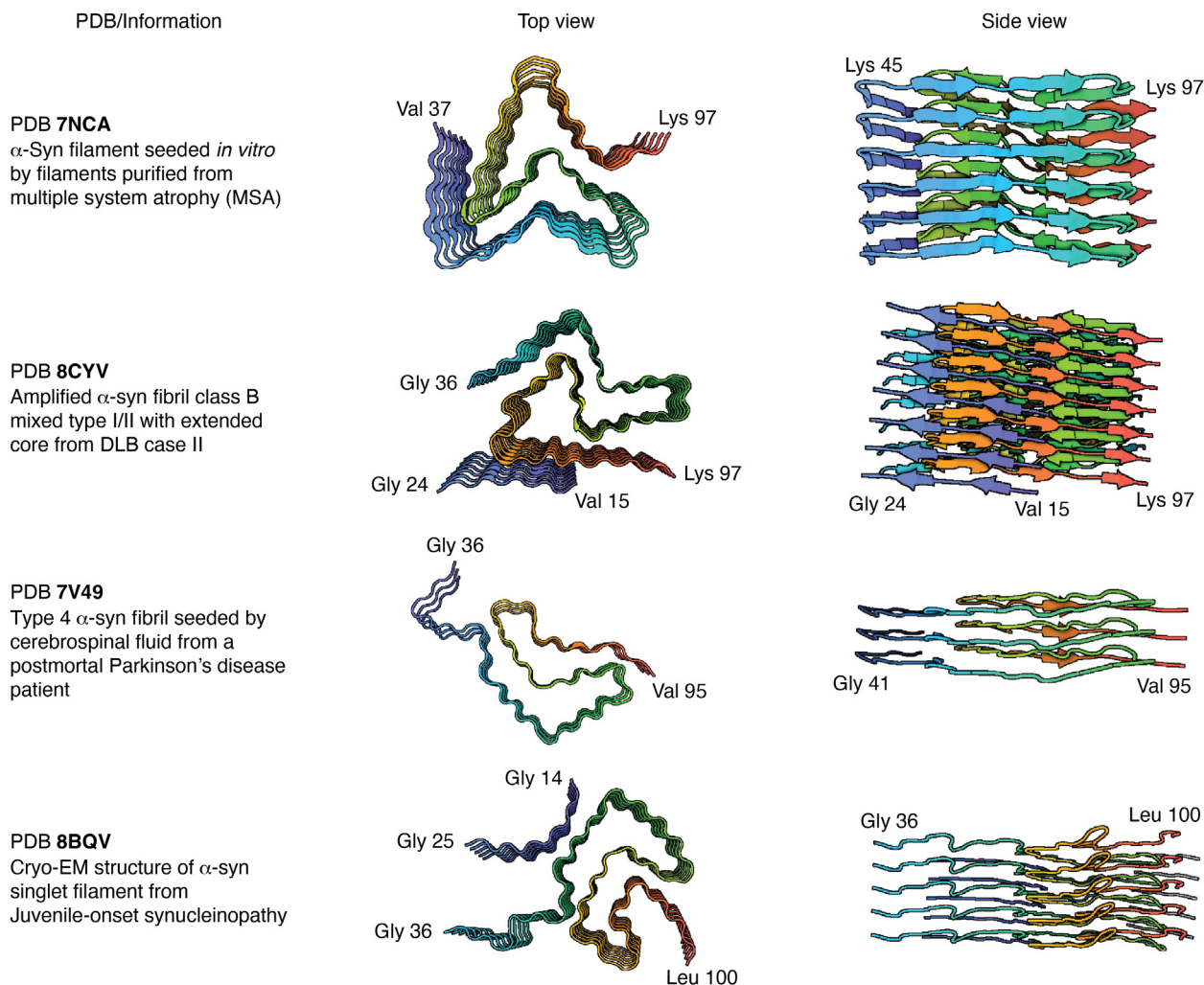


Figure 4. α -syn aggregates in different synucleinopathies. Structures of α -syn fibrils and filaments, seeded *in vitro* or isolated from brains, were resolved using Cryo-EM. The protein data bank (PDB) number for each structure is listed. The structure in 7NCA was assembled using an MSA seed and incubated with recombinant α -syn. This structure was resolved by Lovestam *et al.* (156). 8CYV shows the resolved structure of α -syn fibrils amplified using CSF from a (Dementia with Lewy bodies) DLB case, resolved by Sokratian *et al.* (253). 7V49, shows an α -syn fibril seeded by CSF of a PD patient (254). 8BQV, shows the structure of α -syn filament from juvenile-onset synucleinopathy (255). Although α -syn is the building block of all these aggregates, different diseases and physiological conditions lead to structurally distinct 3D structures. CSF, cerebrospinal fluid; MSA, multiple system atrophy; PD, Parkinson's disease.

LB ultrastructure

For many years, the ultrastructural study of LBs yielded a unanimous observation: a proteinaceous inclusion filled with filaments and aggregated proteins, the most prominent of which was α -syn (157–161). Although some evidence was presented for the incorporation of organelles in LBs (162–164), the article by Shahmoradian *et al.* (60) cemented the following findings and uncovered the true nature of LBs: LBs are composed of an organellar medley filled with membranous and organelle fragments that also contain filamentous structures. Using electron tomography, electron microscopy, and stimulated emission depletion microscopy while taking advantage of the incredibly short postmortem interval before tissue collection, Shahmoradian *et al.* (60) were able to report some of the highest resolution images of LBs to date. This dramatically changed the understanding of the structure of LBs and their origin. As opposed to LBs simply being aggregated proteins within the cytoplasm of neurons, they are now a collection of dysfunctional organelles, membrane fragments, filaments, and cytoskeletal structures. The scope of our understanding regarding the contents of LBs has changed, and with it, the understanding of the underlying mechanisms involved in the formation of LBs should also change.

Formation of LBs

Prior to the findings reported by Soper *et al.* (162) and Shahmoradian *et al.* (60), the process in which LBs were formed was simple: α -syn in the cytosol begins to aggregate, which may be caused by mutations, increased α -syn expression, or some external factor, and the protein starts the process of LB formation; however, with the incorporation of membranes and organelles, the issue of LB formation is now more complex. Still, the majority opinion in the field is that LB formation occurs as a function α -syn fibrillization (165): through its fibrillization, α -syn associates with a multitude of membranes and organelles during its fibrillization, brings together a plethora of cellular components and packages them into an inclusionary body (Fig. 5A). However, that hypothesis would entail that fibril forms of α -syn be readily found in LBs during and after their formation, as they are the driving force behind the formation of LBs. However, Shahmoradian *et al.* (60) pointed out the relative absence of fibrils in their isolated LBs.

In our recent study (148), we characterize the formation of membrane-enclosed LB-like inclusions. There are no signs of fibrils in the formative stages of inclusion formation, and the “mini-inclusions” consist mainly of dark, aberrant lysosomes. Fibrils and filaments are incorporated into these structures at later stages. This finding suggests that α -syn fibrillization is not the driving force behind LB formation. Our membrane-bound inclusions continued to mature by incorporating more and more organelles, including mitochondria, ER, lysosomes, and neurofilaments, resulting in structures similar to those shown by Shahmoradian *et al.* (60). A model of the possible mechanism by which our LB-like inclusions are formed is illustrated in Figure 5B.

In recent years, a few articles have attempted to form LB-like inclusions *in vitro*, and each has been able to recapitulate different characteristics of LBs. Table 3 lists studies that included an ultrastructural survey on their LB-like inclusions.

LBs: neurotoxic or neuroprotective

If LBs are the result of α -syn fibrillization, and α -syn's need to aggregate, then the formation of inclusions such as LBs occurs in a cell-autonomous manner: α -syn aggregates bind and tether organelles as they continue aggregation, which leads to cytosolic LBs (Fig. 5A). This would most likely make LBs neurotoxic, as the sequestration of organelles, inhibiting the transport of enzymes, proteins, and RNAs to and from them, would result in impaired mitochondria releasing cytochrome c and leaky lysosomes releasing hydrolases into the cytosol (166–171). The fibrillization of α -syn would not only cause the formation of inclusions but would lead to Caspase-mediated cell death, necrosis, and apoptosis.

If LBs are formed not by the fibrillization of α -syn but through the cell's attempt to sequester aggregates and damaged organelles, clearing them from the cytosol, this would make LBs neuroprotective, protecting the host neuron from the deleterious releases of damaged organelles, protein misfolding, and aggregation. If LBs are formed through cell-mediated processes, this can only be accomplished through an aggresomal response (172, 173), or through autophagic uptake of damaged organelles and protein aggregates. The former would result in the formation of inclusions with a somewhat impermeable cytoskeletal cage (174). In contrast, the latter would result in the formation of gigantic autophagosomes, that could not degrade the contents in their lumen due to lysosomal defect (175), but could at least recognize and sequester protein aggregates and damaged organelles and clear them from the cytosol (Fig. 5B and 6).

Contents of LBs

A consistent finding in previous research is that LBs are composed of a collection of fibrillar and filamentous materials. With the new ultrastructural information described by Shahmoradian *et al.* (60), organelles such as lysosomes, autolysosomes (double membrane), mitochondria, cytoskeletal/filamentous materials, and vesicles are all present in LBs. Despite their unprecedented insights into LBs, Shahmoradian *et al.* (60) were not the only ones to report the incorporation of cellular organelles in LBs (162–164). In fact, earlier research by Soper *et al.* (162) published some of the clearest electron micrographs of LBs known to date. However, all of these previous studies were limited to the resolution afforded to them by postmortem tissues; hence, much more information about the ultrastructural nature of LBs is yet to be unearthed, using *in vitro* 2D and 3D models or perhaps through culturing of postmortem sections, allowing neurons to recover prior to fixation (176).

LBs and lysosomal proteins

As mentioned previously, the main question in PD and LBD is whether LBs are part of the pathophysiology of these

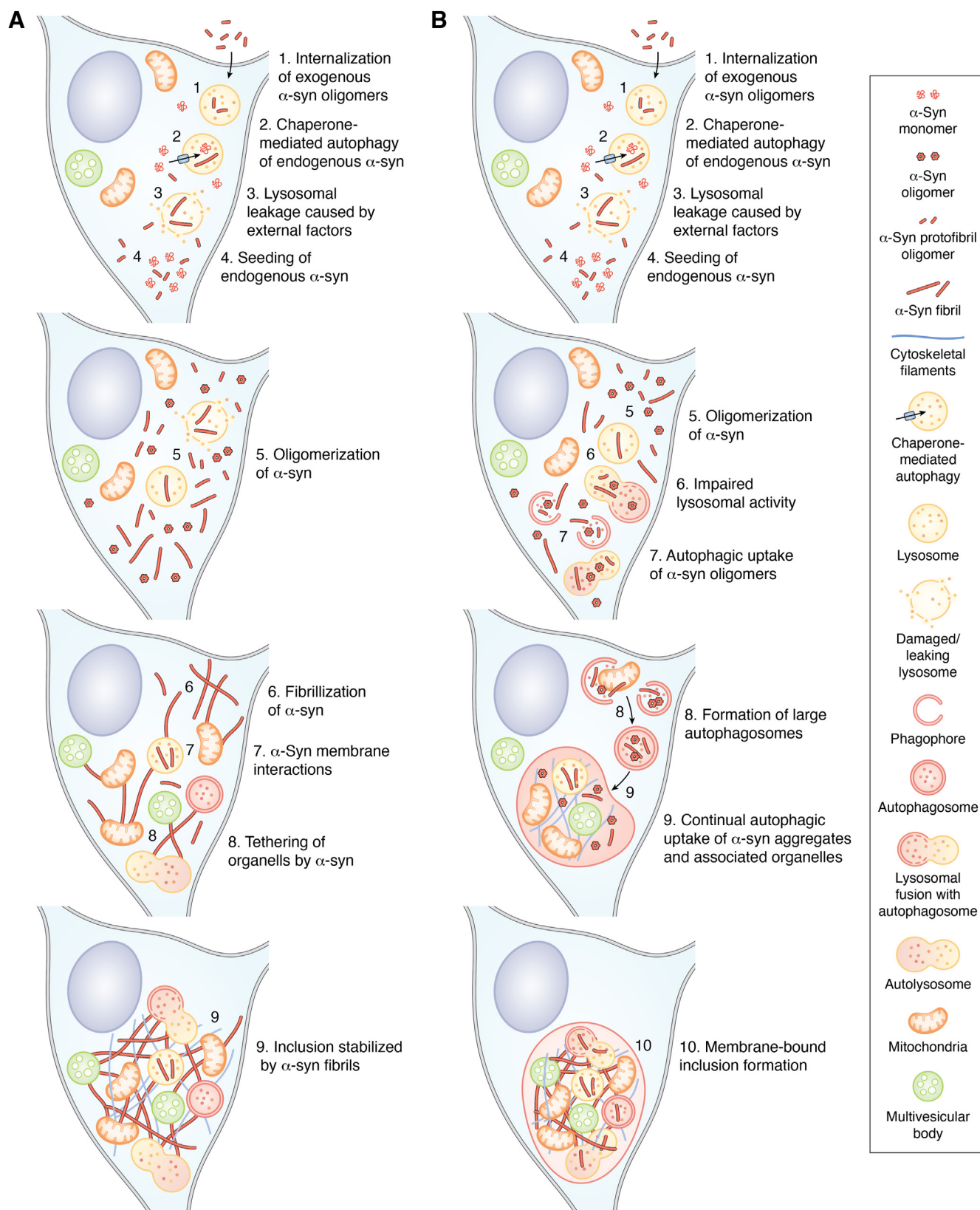


Figure 5. Models for the formation of LBs through fibrillization of α -syn and autophagosomal dysfunction. A, internalization of PFFs and their accumulation in lysosomes (1) is followed by the introduction of cytosolic α -syn into lysosomes through CMA (2), initiating the first steps of endogenous α -syn aggregation. Lysosomal stress caused by α -syn accumulation and aggregation, both from the cytosol and PFFs, and potentially external factors, leads to lysosomal membrane permeabilization (3). Aggregated α -syn oligomers/protofibrils then gain access to the cytosol and seeds the aggregation of endogenous α -syn (4 and 5). Fibrillization of α -syn continues, and α -syn aggregates begin to associate with different organelles through membrane and protein interactions (6–8). α -syn fibrils then act as tethers and form a cytoplasmic inclusion, eventually maturing into LBs (9). B, formation of membrane-bound LBs can occur through the internalization of aggregate forms of α -syn, potentially coupled with an external factor (1). Endogenous α -syn, whose

Table 3***In vitro* models of LB-like inclusions**

Publication	Recapitulated characteristic of LBs	Structure of LB-like inclusions formed	Reference
Yamagishi <i>et al.</i> , 2007	Ubiquitin-positive inclusions	Fibril containing, dense, proteinaceous inclusions	(289)
Bedford <i>et al.</i> , 2008	Ubiquitin, p62, and α -syn positive inclusions.	Inclusions include mitochondria, membrane-bound vesicles, autophagosome like structures, filamentous contents	(290)
Luk <i>et al.</i> , 2009	Ubiquitin, phospho- α -syn and thioflavin S positive inclusions	Proteinaceous aggregate with fibrillar α -syn	(291)
Mahul-Mellier <i>et al.</i> , 2019	Phospho- α -syn positive inclusions	Cytoplasmic organellar medley	(57)
Trinkaus <i>et al.</i> , 2021	α -syn-positive inclusions	α -syn fibrils crisscrossing organelles such as lysosomes, mitochondria, and MVBs.	(292)
Jo <i>et al.</i> , 2021	Halo-like α -syn staining in TH-positive neurons	LB-like core with fibrillar α -syn in the periphery	(293)
Lam <i>et al.</i> , 2022	Ubiquitin, p62, α -syn positive inclusions	An organellar medley, containing vesicles, lysosomes, and filamentous materials.	(39)
Bayati <i>et al.</i> , 2023	Phospho- α -syn positive inclusions with mitochondria, lysosomal contents within	Membrane-bound filamentous structures containing an organellar medley. Organelles within inclusions look dysfunctional.	(148)

diseases or a by-product/symptom of their pathophysiology. Some have suggested that LBs result from an aggresomal response, resulting in the accumulation of proteins (173, 177). This response is indicative of impaired autophagy and dysfunctional degradation by lysosomes. Supporting this hypothesis, others have posed the idea that PD is a type of lysosomal storage disorder (178), and recent research has implicated multiple lysosomal proteins as playing a role in PD, such as TMEM175 (179–182), GBA (183–188), LAMP1 (63, 189, 190), and LAMP2 (46, 47, 191, 192). LAMP2A, the lysosomal membrane protein responsible for CMA (193), is hypothesized to be involved in maintaining α -syn homeostasis (*i.e.*, controlling α -syn concentration) under physiological conditions (194) since impaired CMA and downregulation of LAMP2 has not only been reported in PD (47, 191), but has also been associated with higher α -syn levels in *in vivo* models (46). Moreover, LAMP2 is also responsible for autophagy-lysosomal fusion (195, 196), which is critical for the degradation of contents accumulating in autophagosomes. Lastly, misfolded and aggregated forms of α -syn that form in the cytosol lead to the disruption of the LAMP2A translocation pore, altering its structure and inhibiting CMA (45, 194). Thus, LAMP2 controls α -syn levels in the cell to prevent aggregation; its expression is associated with lower α -syn levels, but α -syn aggregates can disrupt its function, causing a buildup of cytosolic α -syn.

Other lysosomal-related proteins, such as TMEM106B (197), TFEB (198–201), and NRF2 (202–207), have also been implicated in neurodegenerative diseases. TMEM106B, a ubiquitously expressed lysosomal protein highly expressed in the brain, is involved in the cellular trafficking of lysosomes and the maintenance of lysosomal health (*i.e.*, pH and enzymatic activity) (208–210). TFEB is a master transcription factor encoding many proteins associated with lysosomal

activity and biogenesis (211). NRF2 is involved in the cellular inflammasome response and also serves as a transcription factor for LAMP2 (212, 213), making NRF2 expression vital in preserving CMA and autophagosome-lysosomal fusion (202, 214). Its upregulation is associated with cellular antioxidative response. In sum, researchers studying neurodegeneration have come to understand and further explore the role of lysosomes in neurodegenerative diseases, especially in PD and Alzheimer's disease, making the inclusions that form a symptom of the disease and not the disease-causing factor.

LBs and an immune challenge

Finally, it is essential to note the most recent trend in the field of neurodegeneration: neuroinflammation. The involvement of the immune system and microglia in neurodegeneration has been in the zeitgeist (36, 215–219) but has recently received more attention (50, 83, 138, 220–225), with its most staunch support stemming from an article by Matheoud *et al.* (226) in which researchers show that *Pink1*^{−/−} mice, notorious for not recapitulating PD-like neurodegeneration (227–232), experience Parkinsonism and neurodegeneration only after the introduction of gastrointestinal bacteria. Their research suggests that although the *Pink1*^{−/−} mutation (a PD insult) is present in the mice, it takes an external factor, a gastrointestinal infection, and an associated immune system activity to initiate the onset of PD motor symptoms. Other articles have also come to similar conclusions in other neurodegenerative diseases, such as a recent work published by Burberry *et al.* (233) focusing on the immune system's role in amyotrophic lateral sclerosis mice models. The potential involvement of proinflammatory cytokines in neurodegeneration (221, 234), coupled with research further popularizing the possible infiltration of immune cells into the brain (235, 236), has resulted in another viewpoint in neurodegenerative research. This has culminated in researchers'

concentration is controlled by CMA (47), meets internalized PFFs in lysosomes, leading to α -syn aggregation occurring within lysosomes (256) (2). Eventually, mounting lysosomal stress caused by PFFs (and potentially an external factor) leads to lysosomal membrane permeabilization (lysosomal leakage), leading to the leaking of PFFs into the cytosol (3). PFFs then seed the aggregation of endogenous α -syn (4 and 5). Impaired lysosomal activity is then exacerbated by aggregated forms of α -syn blocking the CMA pore, composed mainly of LAMP2A, leading to further accumulation of monomeric α -syn (6). Autophagosomes attempt to clear aggregation and damaged/dysfunctional organelles through autophagic uptake (7). Due to impaired lysosomal activity, contents within autophagosomes are not degraded, and autophagosomes begin expanding in size as the sequestration of aggregates and organelles continues (8 and 9). Eventually, this leads to the formation of membrane-bound LB inclusions (10). CMA, chaperone-mediated autophagy; LB, Lewy body; PFF, preformed fibril.

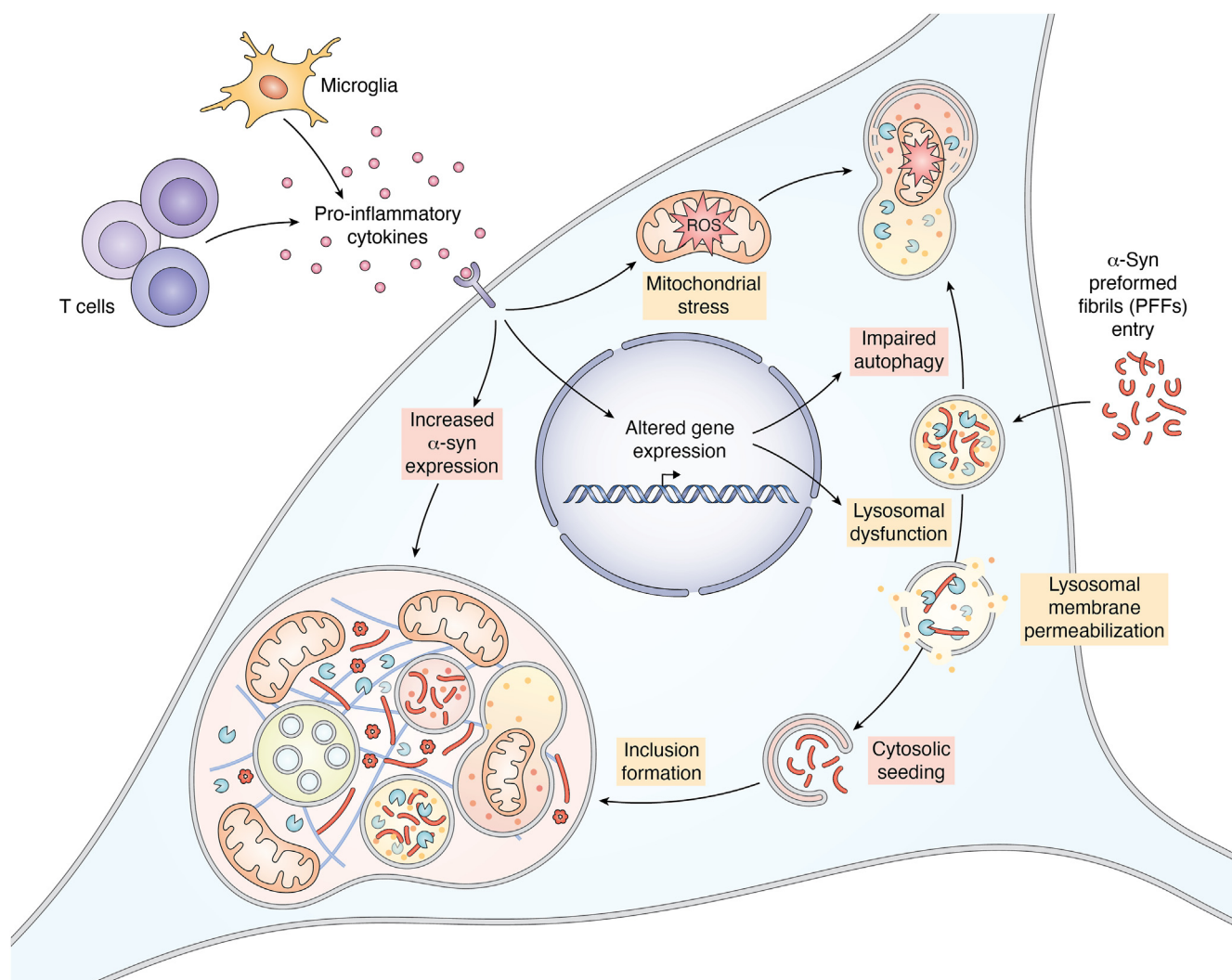


Figure 6. Model of LB-like inclusion formation through a dual hit treatment regime. Internalization of an aggregate form of α -syn, coupled with exposure to proinflammatory cytokines (released by microglia or T-cells), results in two different pathways that work to stress the cell. PFF internalization (a PD insult) results in lysosomal stress, and their occupation of lysosomes results in impaired cell degradative activity. The introduction of proinflammatory cytokines activates immune-related pathways in the cell that alter gene expression. Downregulation of lysosomal proteins results in exacerbating lysosomal dysfunction. Proinflammatory cytokines also lead to the rise of oxidative stress and elevated expression of endogenous α -syn in the cell. The combination of PFF accumulation in lysosomes, altered gene expression, and oxidative stress leads to lysosomal membrane permeabilization and the leaking of PFFs into the cytosol. PFFs gain access to the cytosol and seed the aggregation of endogenous α -syn. Autophagosomes attempt to clear aggregation and dysfunctional organelles, but they cannot due to impaired lysosomal activity. This leads to the formation of membrane-bound LB-like inclusions. LB, Lewy body; PD, Parkinson's disease; PFF, preformed fibril.

increased awareness of microglia and the immune system's role in neurodegeneration.

Data collected in our most recent work on observing the formation of LB-like inclusions *in vitro* (148) involves a dual hit treatment regime, resulting in the formation of LB-like inclusions in DA neurons exposed to IFN- γ . In the absence of this dual hit treatment, internalized exogenous α -syn PFFs remain within lysosomes and multivesicular bodies, maintaining their punctate fluorescence. Neurons that are exposed to IFN- γ , on the other hand, exhibited diffuse PFF fluorescence and underwent the process of forming large PFF-positive aggregates. By EM, these PFF-positive aggregates were full of organelles and filaments, resembling the organelle-filled LBs previously reported (60, 162). A summary of our dual hit hypothesis is illustrated in Figure 6. Our findings cement

the dual hit treatment as one possible pathway for the formation of LBs, the pathological hallmark of PD, but more importantly, our approach can be adapted to view PD as a more multidimensional disorder.

Conclusion: A change in perspective

The mechanistic study of PD pathophysiology, using the many novel experimental models such as induced pluripotent stem cells and organoids, will need to be used to answer many questions about the molecular genesis of PD. For instance, why does Parkinson's disease mainly affect older adults? Why is the prodromal stage of PD such a long process? What is going wrong in cells in the prodromal period to cause prodromal symptoms, including rapid eye movement sleep behavior

disorder and olfactory deficits (237–242), but not result in Parkinson's disease symptoms?

As indicated previously, many factors can contribute to the “perfect storm” that leads to stressed neurons with dysfunctions in the autophagy-lysosomal pathway, but there is one common factor: aging. Aging microglia are more likely to produce proinflammatory cytokines (243). Aging microglia also facilitate the infiltration of T-cells into the brain, perhaps in an effort to restore the lost immune surveillance in the brain. However, like microglia, the aging immune cells are also more likely to release proinflammatory cytokines in response to mild and moderate infections, resulting in a cytokine storm (244–246).

The immune perspective enables further understanding of how aging becomes a risk factor for poor neuronal health, resulting in an environment where neurodegeneration can occur. Neurons will be more likely to produce high levels of reactive oxygen species and will be unable to rid the cell of harmful materials. In addition, organellar biogenesis may be altered, and dysfunctional organelles cannot be degraded. This will push cells into forming inclusions to protect healthy parts of the cells from degeneration. We believe it is the combination of a PD-specific insult (e.g., genetic mutation, α -syn upregulation, and aggregation) along with an external factor, such as toxin exposure, immune system activation, or increased secretion of proinflammatory cytokines, that results in an environment ripe for the onset and occurrence of PD. However, the ease with which this “perfect storm” occurs, and the chances that this dual hit coincide, is increased with aging, resulting in a much higher incidence of PD and other synucleopathies to occur later in life. The mechanistic contributions of all these factors to PD pathophysiology need to be further studied.

It might also be interesting to consider α -syn as an unwilling participant in PD rather than the protein involved in the genesis of the disease. If α -syn is simply an indicator of cellular stress, as posited by recent research (40), then its accumulation and increased expression can happen transiently following neuronal stress, which has been shown previously (49). With its increased expression, there is an ever-increasing chance for it to aggregate due to its NAC/hydrophobic region (247). This results in the formation of aggregates that then lead to the trapping and incorporation of proteins and organelles. Aggregates can alter the shape and function of the LAMP2 pore (45), which is responsible for transporting materials from the cytoplasm to the lysosomes in CMA. Impairment in CMA leads to more accumulation of proteins in the cytoplasm (47, 248), and the altered LAMP structures lead to an inability for lysosomes to fuse with autophagosomes (249). All roads once again lead to a failure in autophagy.

The goal of this review was to present and aid the reader in understanding the multifaceted nature of α -syn: how it seeds the formation of endogenous α -syn and may lead to the formation of inclusions, but also acts as an indicator of cellular stress that is affected by toxins, increasing reactive oxygen species levels, and impairments in autophagy. Our current understanding of the field points to the possibility that immune challenges and the release of proinflammatory cytokines in the brain may be the triggering event that bridges the gap

between prodromal PD and the onset of PD motor symptoms. The involvement of the immune system and proinflammatory cytokines explains the propensity of older populations to develop PD, their susceptibility to cytokine storms in response to infections, and the role that the aging immune cells may play in this process. In short, research into the role of the immune system and microglia may hold the key to our understanding of neurodegenerative diseases' pathophysiology and warrants further investigation.

Acknowledgments—We thank Dr Jo Anne Stratton for her input on our manuscript. Figure 3 was put together using [Biorender.com](https://biorender.com). A. B. is supported by the Fonds de recherche du Québec doctoral award and a studentship from the Parkinson Society of Canada. This manuscript was adapted from the AB's doctoral thesis. This work was supported by financial support from the Canada First Research Excellence Fund, Healthy Brains, Healthy Lives, McGill University, awarded to P. S. M. P. S. M. is a Distinguished James McGill Professor and Fellow of the Royal Society of Canada.

Author contributions—A. B. and P. S. M. writing—original draft; A. B. conceptualization; A. B. visualization; P. S. M. supervision.

Conflict of interest—The authors declare that they have no conflicts of interest with the contents of this article.

Abbreviations—The abbreviations used are: CMA, chaperone-mediated autophagy; CME, clathrin-mediated endocytosis; DA, dopaminergic; GPe, globus pallidus; GPi, globus pallidus internal segment; IFN- γ , interferon-gamma; LB, Lewy body; LBD, Lewy body dementia; NAC, nonamyloid component; PD, Parkinson's disease; PFF, preformed fibril.

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